Dosage Inequity in Pharmaceutical Studies: A Power Analysis*

Replication and Reanalysis

Susan Peck Jan Blackburn Joseph Burks

May 8, 2026

Clinical trials are often powered to detect overall treatment effects, but not treatment effect differences within demographic subgroups. This paper reanalyzes the ACTG 320 HIV clinical trial to evaluate whether sex and race/ethnicity subgroups had adequate statistical power to support subgroup-specific conclusions. Using Cox proportional hazards models, Kaplan–Meier curves, likelihood ratio tests, and Schoenfeld-based power calculations, we replicate the original overall treatment finding and then assess subgroup treatment effects. Indinavir-containing therapy reduced the hazard of AIDS progression or death by approximately 48% in the full sample, but most subgroup analyses were underpowered because statistical information in survival analysis depends primarily on observed events rather than enrollment counts. The female subgroup and overall sample exceeded 80% power, while the male, Hispanic, Asian/Pacific Islander, and Other subgroups had too few events to support reliable inference. These results show that demographic representation alone does not guarantee inferential integrity and that future trials should use event-driven subgroup targets when subgroup-specific inference is a design objective.

1 Introduction

Clinical trials are used for evaluating treatment efficacy, but they are typically designed and powered to detect treatment effects in the overall enrolled population. This means that demographic subgroups such as women and minorities are often enrolled in numbers too small to support reliable inference about whether a treatment works equally well, or works at all, for those groups. When a trial lacks the power to detect subgroup differences, the absence of a significant finding cannot conclusively be interpreted as evidence that the treatment effect

*Project repository available at: https://github.com/susanpeck/Math261BExperimentalDesignProject is the same across all subgroups, and yet, treatment guidelines are often generalized to all demographic groups.

The ACTG 320 trial (Hammer et al. 1997) is an HIV clinical trial that provides an instructive case study for examining these limitations. ACTG 320 randomized 1,156 patients with advanced HIV disease to either a standard two-drug nucleoside regimen or a three-drug regimen adding the protease inhibitor indinavir. The trial was powered for an overall treatment comparison and successfully demonstrated that indinavir containing therapy reduced the hazard of AIDS progression or death by approximately 50%. However, enrollment was highly imbalanced across demographic groups, with female participants comprising over 80% of the sample and several subgroups represented by only a handful of participants and even fewer observed events.

The paper is going to investigate if the demographic subgroups were adequately powered to detect treatment effects, and was indinavir's benefit consistent across subgroups. In addition, what enrollment would be required in a future study to achieve adequate power for all demographic subgroups? In studies like ACTG 320 survival analysis is driven by the number of observed events, not enrollment counts alone. A subgroup can be proportionally represented in a trial while still contributing too few events to support stable hazard ratio estimation or meaningful hypothesis testing.

The original ACTG 320 survival analysis will be replicated using Cox proportional hazards models and Kaplan–Meier curves and then extends it through a systematic investigation of sex and race/ethnicity subgroup treatment effects and prospective power analyses. The key finding is that while the overall treatment effect and the female subgroup were adequately powered, the male, Asian/Pacific Islander, Hispanic, and Other race subgroups were not. The paper will describe relevant literature on demographic representation in clinical trials (Section 2). It will then outline the ACTG 320 study design (Section 3) and present the statistical analysis plan and power analysis framework (Section 5). The reanalysis results are presented in (Section 6), and findings, limitations, and implications for future trial design are discussed in (Section 7).

2 Literature Review

The underrepresentation of women and racial and ethnic minorities in clinical trials has been documented across decades of pharmaceutical research. Three studies are briefly described here.

Poon et al. (2013) examined women's participation and the reporting of sex-stratified analyses in 156 late-phase trials supporting FDA approval from 2000 to 2009. Women made up 45% of participants on average, with the greatest imbalances in cardiovascular and infectious disease trials. Even when women were enrolled in adequate numbers, sex-specific results were rarely reported. Fewer than one-third of trials provided subgroup results by sex, suggesting that enrollment alone is not sufficient to support meaningful conclusions about treatment effects in women.

Khan et al. (2020) analyzed late-phase cardiometabolic drug trials submitted to the FDA between 2008 and 2017, focusing on the enrollment of racial and ethnic minorities. Minority enrollment improved only modestly over the decade and remained well below the representation of those groups among patients with the relevant diseases. Even when minority participants were enrolled, absolute numbers were often too small to support meaningful subgroup analyses. The authors call for prospective enrollment targets for underrepresented groups as a design requirement rather than an afterthought.

Eshera et al. (2015) documented the same pattern across all therapeutic areas in late-phase trials from 2005 to 2012, finding persistent underrepresentation of older adults, women, and racial minorities relative to disease burden in the general population. The study highlights that this is not a problem confined to any single disease area but reflects broader tendencies in how late-phase trials are designed and conducted.

Together, these studies establish that underrepresentation in late-phase trials is a widespread and persistent problem with real consequences for whether approved drugs are proven safe and effective across diverse patient populations

3 Study Design

The ACTG 320 trial (Hammer et al. 1997) was a multicenter, randomized, double blind, placebo-controlled clinical trial evaluating whether adding the protease inhibitor indinavir to a standard two-drug nucleoside analogue regimen improved outcomes in HIV-infected patients with CD4 white blood cell counts of 200 cells/mm^3 or fewer. A total of 1,156 adults were enrolled across multiple clinical sites. Participants were split into two blocks based on their CD4 count measuring disease severity (below 50 or between 51-200 cells/mm^3), and then randomly assigned to control or treatment.

The primary response variable is the number of days from randomization to progression to an AIDS defining illness or death, whichever occurred first. This paper additionally examines participant sex (female, male) and race/ethnicity (White non-Hispanic, Black non-Hispanic, Hispanic, Asian/Pacific Islander, Other) as potential effect modifiers. All models adjust for the following baseline covariates: age at the start of the trial, baseline CD4 cell count, Karnofsky performance score from 0 to 100 indicating how well a patient can carry out normal daily activities, months of prior zidovudine (ZVD) use, and a binary indicating which disease severity block for randomization.

The design had several strengths. Randomization reduced confounding by assigning participants to treatment groups by chance, so that both known and unknown factors that could affect the outcome were distributed roughly equally across groups. Double blinding prevented participants and researchers from knowing treatment assignments, and blocking ensured that disease severity was evenly distributed across treatment groups. The relatively large overall enrollment gave the study enough power to detect the primary treatment effect.

A key limitation for this reanalysis is that the trial was not designed to support subgroup inference. No minimum sample size or event count targets were specified for demographic subgroups, limiting the ability to draw reliable conclusions about treatment effect heterogeneity.

4 Dataset

The dataset used in this analysis is the ACTG 320 dataset, obtained from the `GLDreg` R library (Su 2023). The data contain participant level records for 1,151 enrolled patients and were originally collected as part of a randomized clinical trial described in Hammer et al. (1997). The original trial enrolled 1,156 participants, and the five-record discrepancy between the published enrollment count and the available dataset is unexplained in the public data release.

The observational unit is the individual trial participant. Each record contains one row per participant with the following variables relevant to this analysis:

- `time`: days from randomization to the primary endpoint event or censoring
- `censor`: event indicator (1 = AIDS-defining event or death observed, 0 = censored, meaning no defining event or death at last observation)
- `tx`: treatment assignment (1 = indinavir regimen, 0 = control)
- `sex`: participant sex (0 = female, 1 = male)
- `raceth`: race/ethnicity coded as a five-level factor (1 = White non-Hispanic, 2 = Black non-Hispanic, 3 = Hispanic, 4 = Asian/Pacific Islander, 5 = Other)
- `cd4`: baseline CD4 cell count (cells/mm^3) measured at enrollment
- `age`: participant age in years at enrollment
- `karnof`: Karnofsky performance score, a measure of functional status from 0 to 100, with higher scores indicating better function
- `priorzdv`: months of prior zidovudine (ZDV) use before enrollment
- `strat2`: CD4 count indicator used for randomization (0 = CD4 < 50 cells/mm^3, 1 = CD4 51–200 cells/mm^3)

A total of 96 participants experienced the primary endpoint during follow-up, yielding an overall event rate of approximately 8.3%. This means that the statistical power for subgroup analyses is driven primarily by the small absolute number of events within each subgroup rather than by enrollment counts.

5 Statistical Analysis Plan and Power Analysis

Clinical trials evaluating time-to-event outcomes require statistical methods that properly account for censoring and the timing of events. In the context of the ACTG 320 study, the primary endpoint is the time to progression to AIDS or death, making survival analysis the appropriate framework. The section will start with a review of key concepts in **survival**

analysis and progress to regression model formulations, hypothesis testing, and power analysis definitions.

Let T denote a non-negative random variable representing the time to the event of interest. The survival function is defined as \$ S(t) = P(T > t)\$. This quantity represents the probability that an individual remains event free beyond time t and provides the basis for survival and hazard analysis. The **hazard function** is defined as

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T > t)}{\Delta t} \quad (1)$$

where Δt is a small time interval, and $h(t)$ represents the instantaneous event rate at time t conditional on survival up to time t (Moore 2016a). The hazard function represents how quickly the survival probability declines over time. Higher values of $h(t)$ correspond to a greater risk of the event occurring immediately after time t, while lower values indicate slower event occurrence.

The **Cox proportional hazards model** estimates the association between covariates and the event outcomes in time-to-event clinical studies. The model is semi-parametric since it doesn't require specifying the baseline hazard function. For subject i with covariate vector X_i, the Cox model is

$$h_i(t \mid \mathbf{x}_i) = h_0(t) \exp(\mathbf{x}_i^\top \beta),$$

where $h_0(t)$ is the unspecified baseline hazard function and β is a vector of regression coefficients. In the Cox model, comparisons between groups are made through ratios of hazard functions, allowing treatment effects to be interpreted as relative differences in instantaneous risk over time (Moore 2016a). The hazard ratio (HR) compares the hazard function between two groups

$$HR = \frac{h_T(t)}{h_C(t)}$$

where $h_T(t)$ is the hazard function for the treatment group and $h_C(t)$ hazard function for the control group. An HR less than one indicates the treatment reduces the instantaneous risk and is more effective than the control group. The Cox model assumes a constant hazard ratio over all time intervals.

The Cox proportional hazards model is estimated using partial likelihood estimation to account for censoring of the data (Moore 2016b). The partial likelihood is converted to the log-likelihood which is maximized to obtain the estimate $\hat{\beta}$.

$$L(\beta) = \prod_{i:\delta_i=1} \frac{\exp\left(X_i^\top \beta\right)}{\sum_{j \in R(t_i)} \exp\left(X_j^\top \beta\right)}.$$

The likelihood ratio test is generally preferred in survival analysis for evaluating potential changes in interaction effects (Moore 2016b). To test a hypothesis about a subset of parameters, two models are considered: - a **reduced model**, which imposes the null hypothesis constraints

- a **full model**, which includes all parameters of interest The test compares how well each model fits the data using a measure called the likelihood. If the full model fits the data much better than the reduced model, the test returns a small p-value, indicating the extra parameters are worth including.

The **overall treatment effect** may be evaluated using a Cox proportional hazards model of the form

$$h_i(t) = h_0(t) \exp\left(\beta_1 T_i + \beta_2^\top Z_i\right)$$

where T_i is the treatment indicator and Z_i is a vector of additional baseline covariates. The coefficient β_1 represents the log hazard ratio for the primary treatment effect after adjusting for the covariates in Z_i. The treatment hazard ratio is $HR_T = \exp(\beta_1)$. The null hypothesis for the overall treatment effect is

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_A : \beta_1 \neq 0$$
$$\text{or equivalently}$$
$$H_0 : HR_T = 1 \quad \text{vs.} \quad H_A : HR_T \neq 1$$

To evaluate whether the treatment effect **differs by sex**, the regression model is expanded to include a gender indicator and a treatment-by-gender interaction term where female was the reference group:

$$h_i(t) = h_0(t) \exp\left(\beta_1 T_i + \beta_2 G_i + \beta_3 T_i G_i + \beta_4^\top Z_i\right)$$

where G_i is a binary gender indicator and Z_i represents additional covariates, β_1 is the treatment effect for the reference gender group, β_2 is the baseline difference in hazard between gender groups among participants in the control treatment group, and β_3 is the treatment-by-gender interaction effect.

The hypothesis test for whether the treatment effect differs by gender is

$$H_0 : \beta_3 = 0 \quad \text{vs.} \quad H_A : \beta_3 \neq 0$$
$$\text{or equivalently}$$
$$H_0 : exp(\beta_3) = 1 \quad \text{vs.} \quad H_A : exp(\beta_3) \neq 1$$

The same interaction framework can be extended to evaluate whether treatment effects **differ across ethnic groups**. Using White non-Hispanic participants as the reference group, the model is defined as

$$\begin{aligned} h_i(t) = h_0(t) \exp\big(&\beta_1 T_i + \beta_2 Black_i + \beta_3 Hispanic_i + \beta_4 AsianPI_i + \beta_5 Other_i \\ &+ \beta_6 T_i Black_i + \beta_7 T_i Hispanic_i + \beta_8 T_i AsianPI_i + \beta_9 T_i Other_i + \beta_{10}^\top Z_i\big) \end{aligned}$$

In this model, β_1 is the treatment effect for White non-Hispanic participants. The coefficients β_2 through β_5 represent baseline hazard differences between each ethnic subgroup and the White non-Hispanic reference group among participants in the control treatment group.

The interaction coefficients β_6 through β_9 measure whether the treatment effect differs between each ethnic subgroup and the White non-Hispanic reference group. The vector Z_i includes additional baseline covariates.

The joint hypothesis test for whether treatment effects differ across ethnic groups is

$$H_0 : \beta_6 = \beta_7 = \beta_8 = \beta_9 = 0 \quad vs. \quad H_A : \text{At least one coefficient} \neq 0$$

If the null hypothesis is rejected, there is evidence that the treatment effect differs for at least one ethnic subgroup relative to the White non-Hispanic reference group.

For a two-group survival comparison, **power** can be estimated directly when the number of observed or anticipated events is known. For a two-sided test at significance level α, the approximate power is

$$\text{Power} = \Phi\left(\sqrt{Dp(1-p)}\,|\log(HR)| - z_{1-\alpha/2}\right),$$

where D is the total number of observed or anticipated events, p is the proportion assigned to the treatment group, $1-p$ is the proportion assigned to the control group, HR is the target hazard ratio comparing treatment with control (Schoenfeld 1981).

Detecting differences in treatment effects between subgroups requires more data than detecting the overall treatment effect. This is because a subgroup interaction effect is estimated by comparing two separate estimates. Specifically, the treatment effect in one subgroup is compared to the treatment effect in another subgroup, which uses less information than the overall analysis. As a result, substantially larger event counts are typically needed to detect whether treatment effects differ across subgroups (Schoenfeld 1981).

For a two-sided interaction test at significance level α, the approximate power is

$$\text{Power} = \Phi\left(\sqrt{D\,p_T(1-p_T)\,p_G(1-p_G)}\,|\log(HR_{\text{int}})| - z_{1-\alpha/2}\right),$$

where p_T is the proportion assigned to the treatment group, p_G is the proportion of participants in one of the subgroup categories (for example, one gender subgroup or one ethnic subgroup) (Schoenfeld 1981).

The interaction hazard ratio quantifies the relative difference between subgroup-specific treatment effects. For example, $HR_{\text{int}} = \frac{HR_1}{HR_0}$.

6 Reanalysis

This analysis is organized into four sections: whole group, sex subgroup, race subgroup, and power analysis. We use R (R Core Team 2026) and the package `survival` (Therneau 2026) to perform the analysis and generate exhibits in this paper.

6.1 Whole Group

We begin by replicating the primary finding from Hammer et al. (1997) using the full study sample. The Kaplan-Meier curves in Figure 1 show the probability of remaining event-free over time for each treatment group. The two curves separate early and remain apart, suggesting that indinavir reduced the hazard of AIDS progression or death.

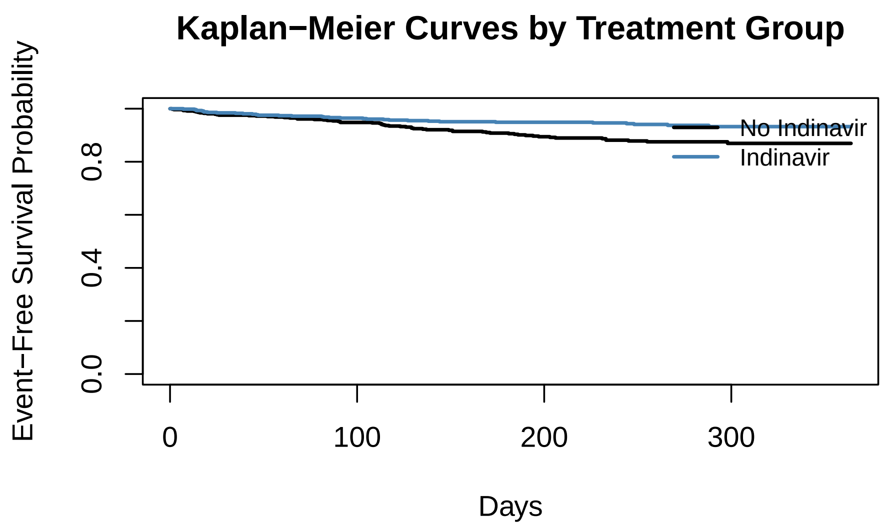

Figure 1: Kaplan-Meier survival curves for Indinavir versus No Indinavir treatment groups.

To estimate the treatment effect while adjusting for patient characteristics, we fit a Cox proportional hazards model. The model includes treatment assignment, baseline CD4 count, age, Karnofsky performance score, months of prior ZDV use, and CD4 count stratum. The full regression results are shown in Table 1.

Table 1: Whole-Group Baseline Cox Model — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment: Indinavir vs. No Indinavir	0.517	0.339	0.788	0.002
Age	1.022	1.000	1.045	0.054
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.926	0.970	0.000
Prior ZDV Use	1.000	0.992	1.007	0.925
CD4 Stratum	1.012	0.517	1.984	0.971

The estimated hazard ratio for treatment is HR = 0.517. This means that at any point in time during follow-up, patients receiving indinavir had approximately half the hazard of experiencing an AIDS-defining event or death compared to the control group, after adjusting for other covariates. This is consistent with the original findings of Hammer et al. (1997). A log-rank test confirms overall treatment group separation, with a chi-square statistic of 10.545 on 1 degree of freedom (p = 0.0012), indicating that the difference in event timing between the two treatment groups is statistically significant.

6.2 Sex Subgroup

To examine whether the treatment effect differed between men and women, we fit two models. The first is a baseline model that includes sex as a fixed covariate alongside treatment. The second is a full model that adds a treatment-by-sex interaction term. A likelihood ratio test compares the two models. We then fit separate Cox models for females and males to estimate sex-specific hazard ratios.

The sex baseline and full interaction models are reported in Appendix Tables A1 and A2. The baseline treatment hazard ratio remains approximately unchanged at 0.517, and the coefficient for sex is not statistically significant, indicating that sex does not meaningfully confound the overall treatment effect. In the full interaction model, the hazard ratio for treatment among females, the reference group, is 0.46. The interaction hazard ratio is 2.034, but the likelihood ratio test yields a p-value of 0.2187, so adding the interaction term does not significantly improve model fit. We cannot conclude that sex modifies the treatment effect.

The sex-specific models yield a hazard ratio of 0.459 for female participants and 1.101 for male participants. With only 15 events among male participants across 200 enrolled individuals, the Cox model for males is severely underpowered, and the estimated treatment effect should be interpreted with caution.

6.3 Race/Ethnicity Subgroup

To examine whether the treatment effect differed across race and ethnicity groups, we fit a baseline model that includes race/ethnicity as a fixed covariate and a full model that adds treatment-by-race/ethnicity interaction terms. Race/ethnicity is coded as a five-level factor with White non-Hispanic participants as the reference group. We then fit separate Cox models within each race/ethnicity group to estimate group-specific hazard ratios.

The baseline model yields a treatment hazard ratio of 0.526, indicating that after adjusting for race/ethnicity, patients receiving indinavir had approximately 47% lower hazard of AIDS-defining events or death. The full race/ethnicity interaction model is reported in Appendix Table A3. The interaction terms for Black, Hispanic, and Other are not statistically significant, providing no clear evidence that the treatment effect differs across these groups. The interaction term for the Asian/Pacific Islander group has a p-value of 0.033, but the confidence interval is extremely wide because there were only 3 events in that subgroup. The likelihood ratio test gives a chi-square statistic of 8.326 with 4 degrees of freedom (p = 0.0803), so we cannot conclude that race/ethnicity modifies the treatment effect.

6.4 Summary of All Effect Sizes

Table 2 presents all hazard ratio estimates across the whole group, sex, and race/ethnicity analyses. Estimates are stable for the overall group and larger subgroups but become unreliable as event counts fall. For Asian/PI and Other, the estimates are essentially uninformative. An HR less than 1 indicates that indinavir-containing therapy reduces the hazard of AIDS-defining events or death relative to nucleoside-only therapy.

Table 2: Summary of Treatment Effect Sizes Across All Analyses

Analysis	Model	Subgroup	n	Events	HR	95% CI Lower	95% CI Upper
Whole Group	Baseline	All	1151	96	0.517	0.339	0.788
Sex	Baseline (pooled)	All	1151	96	0.517	0.339	0.789
Sex	Stratum-specific	Female	951	81	0.459	0.288	0.732
Sex	Stratum-specific	Male	200	15	1.101	0.393	3.087
Race	Baseline (pooled)	All	1151	96	0.526	0.344	0.805
Race	Stratum-specific	White	596	51	0.539	0.303	0.960
Race	Stratum-specific	Black	327	21	0.287	0.104	0.794
Race	Stratum-specific	Hispanic	203	20	0.628	0.246	1.602
Race	Stratum-specific	Asian/PI	14	3	0.068	0.000	Inf
Race	Stratum-specific	Other	11	1	237.311	0.000	Inf

6.5 Power Analysis

Before estimating sample sizes for future studies, we calculated the observed power of the ACTG 320 analyses for the overall population and each subgroup. Power was estimated assuming equal allocation between treatment and control groups. To account for multiple hypothesis testing, we used a Bonferroni-adjusted significance level of $\alpha = 0.025$ for each interaction test. This adjustment controls the family-wise error rate across the two primary subgroup comparisons (sex and race/ethnicity), ensuring that the overall probability of a Type I error remains bounded at 0.05.

The observed power estimates are shown in Table 3. Only the overall group and the female subgroup exceeded the 80% power threshold. All other subgroups fell below it, and for the male, Asian/Pacific Islander, and Other subgroups, the underlying hazard ratio estimates are themselves too unstable to produce meaningful power calculations.

Table 3: Initial Observed Power by Overall Group and Subgroup

Analysis	Subgroup	n	Events	Observed HR	Event Rate	Observed Power	Interpretation
Whole Group	All	1151	96	0.517	0.083	0.839	At or above 80%
Sex	Female	951	81	0.459	0.085	0.897	At or above 80%
Sex	Male	200	15	1.101	0.075	0.020	Below 80%
Race/Ethnicity	White	596	51	0.539	0.086	0.486	Below 80%
Race/Ethnicity	Black	327	21	0.287	0.064	0.732	Below 80%
Race/Ethnicity	Hispanic	203	20	0.628	0.099	0.115	Below 80%
Race/Ethnicity	Asian/PI	14	3	0.068	0.214	0.535	Below 80%
Race/Ethnicity	Other	11	1	237.311	0.091	0.689	Below 80%

Detailed sample size calculations are provided in Appendix Tables A4–A6. For subgroups with more stable hazard ratio estimates, required sample sizes were calculated at 80%, 85%, 90%, and 95% power. For male, Asian/PI, and Other subgroups, where the observed hazard ratios were unstable, required sample sizes were calculated across assumed hazard ratios from 0.9 to 1.1. The Hispanic subgroup stands out as the only stable subgroup that was underpowered in ACTG 320, requiring approximately 447 participants to achieve 80% power compared to the 203 enrolled.

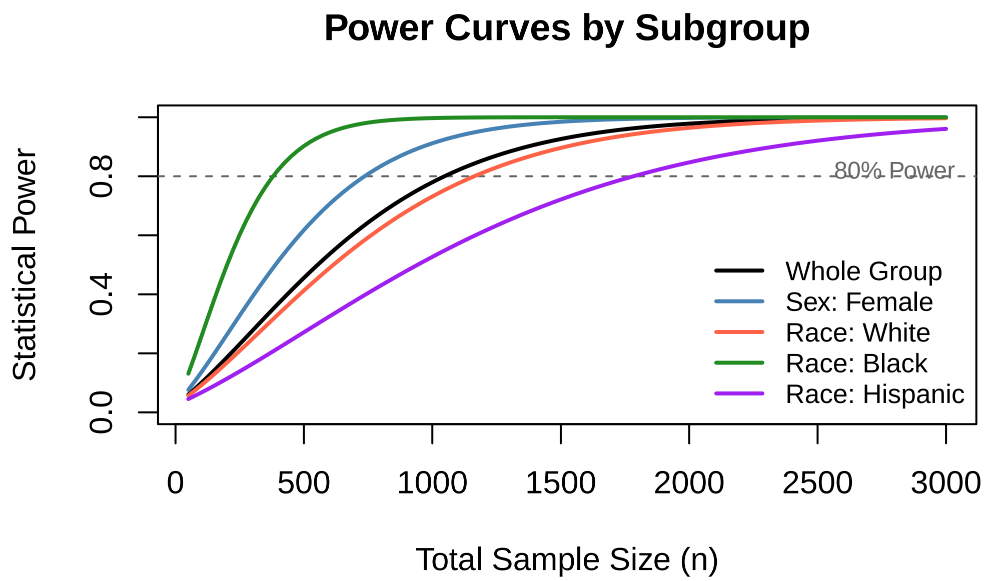

Figure 2: Power curves showing achieved power as a function of sample size for each reliable subgroup, with the 80% power threshold shown.

Table 4: Adequacy of ACTG 320 Sample Sizes for 80% Power

Subgroup	Observed HR	ACTG 320 n	Required n for 80% Power	ACTG 320 Status
Whole Group	0.517	1151	264	Adequate
Sex: Female	0.459	951	188	Adequate
Race: White	0.539	596	293	Adequate
Race: Black	0.287	327	109	Adequate
Race: Hispanic	0.628	203	447	Underpowered

7 Discussion

This paper reanalyzed the ACTG 320 clinical trial (Hammer et al. 1997) to evaluate how demographic subgroup representation and observed event counts affect statistical power in time-to-event analyses. The result for the primary treatment was successfully replicated showing that indinavir-containing combination therapy reduced the hazard of progression to AIDS or death by approximately 48% relative to the control regimen (HR = 0.517). This result was stable across all baseline model specifications and was statistically significant (log-rank p-value = 0.0012), confirming that the overall study had adequate power for its intended purpose.

Subgroup analyses detected no statistically significant treatment-by-sex or treatment-by-race/ethnicity interaction effects. However, this result is not evidence of constant treatment effects across subgroups. Analysis of most subgroups were never in a position to detect meaningful differences as exemplified by the male subgroup. Only 15 events across 200 enrolled participants were documented, leaving the subgroup specific Cox model severely underpowered and estimates unreliable. The Asian/Pacific Islander and Other race subgroups contributed only 3 and 1 events, respectively, making their estimates entirely unstable. Even among the more stable subgroups, only the overall group and the female subgroup exceeded the 80% power threshold at the observed effect sizes. The Hispanic subgroup enrolled 203 participants but remained underpowered because its low event count (20 events) was insufficient to detect a moderate treatment effect.

The central lesson of the power analysis is that observed events, not enrollment, determine power in survival studies. This distinction is not always addressed in the clinical trial literature since the focus is often on enrollment proportions. A subgroup that is adequately represented in a trial may still contribute too few events to support any meaningful conclusion about treatment effects by subgroup. For underpowered subgroups, achieving 80% power often requires enrollment increases that are prohibitively large for a single trial.

7.1 Limitations

The analysis in this paper relies on secondary use of a publicly available dataset and can only use variables provided. Secondary endpoints from the original trial, CD4 cell count trajectories, and HIV-1 RNA concentrations, are not available, limiting reanalysis to the primary time-to-event outcome. Second, the total number of primary endpoint events was small (96 across 1,151 participants), which constrains the precision of all subgroup analyses. Lastly, the power calculations rely on asymptotic approximations under the Cox proportional hazards framework and assume that hazard ratios remain proportional over the follow-up period. This assumption may not hold in all subgroups, given the sparse event counts.

7.2 Implications and Future Research

The findings of this paper have direct implications for the design of future clinical trials where subgroup inference is an objective. Trials powered only for overall treatment comparisons are structurally unable to detect differential treatment responses in underrepresented demographic groups because the study lacks the power to detect them. This issue is especially important for groups that are underrepresented in clinical trials, since they may respond differently to treatments because of biological, medical, or social factors. As (Khan et al. 2020) and (Poon et al. 2013) document, this pattern is a persistent issue in clinical trials across most therapeutic areas.

Prospective enrollment targets for key demographic subgroups should be set based on anticipated event rates and minimum detectable effect sizes rather than just proportional representation. This would ensure that subgroup analyses are powered to detect meaningful differences in treatment effect. Extending follow-up duration can increase the number of events documented without requiring additional enrollment, which may be a cost-effective strategy to improve subgroup power.

Another option is to adopt adaptive event-driven designs that trigger subgroup analyses only after reaching subgroup-specific event thresholds. For situations where no single trial can achieve adequate subgroup enrollment, Bayesian hierarchical models that borrow information across subgroups under assumptions of partial exchangeability may provide a solution(Hannigan et al. 2022). Future work should be performed to evaluate the performance of these approaches to determine if they can address the challenges illustrated in this paper.

8 Appendix: Supplemental Tables

8.1 Supplemental Sex Model Tables

Table 5: Appendix Table A1. Sex Baseline Cox Model — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.517	0.339	0.789	0.002
Sex (Male vs Female)	1.122	0.643	1.957	0.685
Age	1.023	1.000	1.046	0.050
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.926	0.970	0.000
Prior ZDV Use	1.000	0.992	1.007	0.926
CD4 Stratum	1.013	0.517	1.985	0.970

Table 6: Appendix Table A2. Sex Full Cox Model with Interaction — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.460	0.288	0.734	0.001
Sex (Male vs Female)	0.854	0.405	1.801	0.679
Age	1.023	1.000	1.046	0.050
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.947	0.925	0.969	0.000
Prior ZDV Use	0.999	0.992	1.007	0.865
CD4 Stratum	1.011	0.516	1.978	0.975
Treatment x Sex	2.034	0.663	6.238	0.214

8.2 Supplemental Race/Ethnicity Model Table

Table 7: Appendix Table A3. Race Full Cox Model with Interaction — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.534	0.302	0.946	0.032
Race: Black (vs White)	0.794	0.432	1.457	0.456
Race: Hispanic (vs White)	1.002	0.524	1.914	0.995
Race: Asian/PI (vs White)	0.860	0.117	6.317	0.882
Race: Other (vs White)	2.831	0.378	21.227	0.311
Age	1.024	1.001	1.048	0.038

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.927	0.971	0.000
Prior ZDV Use	0.997	0.989	1.005	0.467
CD4 Stratum	1.010	0.513	1.986	0.978
Treatment x Black	0.591	0.185	1.883	0.373
Treatment x Hispanic	1.249	0.422	3.698	0.687
Treatment x Asian/PI	15.514	1.244	193.477	0.033
Treatment x Other	0.000	0.000	Inf	0.993

8.3 Supplemental Power Analysis Tables

Table 8: Appendix Table A4. Required Sample Sizes by Subgroup and Power Level ($\alpha = 0.025$)

Subgroup	Observed HR	Obs. n	Obs. Events	Event Rate	n (80%)	n (85%)	n (90%)	n (95%)
Whole Group	0.517	1151	96	0.083	264	300	348	420
Sex: Female	0.459	951	81	0.085	188	212	247	294
Race: White	0.539	596	51	0.086	293	339	386	468
Race: Black	0.287	327	21	0.064	109	109	125	156
Race: Hispanic	0.628	203	20	0.099	447	508	589	711

Table 9: Appendix Table A5. Required Sample Sizes for 80% Power Across Assumed Hazard Ratios

Subgroup	Assumed HR	Observed Event Rate	Required n for 80% Power
Sex: Male	0.9	0.075	11427
Sex: Male	1.0	0.075	Inf
Sex: Male	1.1	0.075	13960
Race: Asian/PI	0.9	0.214	4000
Race: Asian/PI	1.0	0.214	Inf
Race: Asian/PI	1.1	0.214	4886
Race: Other	0.9	0.091	9427
Race: Other	1.0	0.091	Inf

Subgroup	Assumed HR	Observed Event Rate	Required n for 80% Power
Race: Other	1.1	0.091	11517

Table 10: Appendix Table A6. Required Events and Sample Sizes for 80% Power

Subgroup	Observed HR	Event Rate	Events Required for 80% Power	Required n for 80% Power
Whole Group	0.517	0.083	22	264
Sex: Female	0.459	0.085	16	188
Race: White	0.539	0.086	25	293
Race: Black	0.287	0.064	7	109
Race: Hispanic	0.628	0.099	44	447

References

Eshera, N., H. Itana, L. Zhang, G. Soon, and E. Fadiran. 2015. "Demographics of Clinical Trial Participants in Pivotal Trials for New Molecular Entity Drugs." *The American Journal of Therapeutics* 22 (6): 435–55.

Hammer, Scott M., Kathleen E. Squires, Michael D. Hughes, Janet M. Grimes, Lisa M. Demeter, Judith S. Currier, Joseph J. Eron, et al. 1997. "A Controlled Trial of Two Nucleoside Analogues Plus Indinavir in Persons with Human Immunodeficiency Virus Infection and CD4 Cell Counts of 200 Per Cubic Millimeter or Less." *New England Journal of Medicine* 337 (11): 725–33. https://doi.org/10.1056/NEJM199709113371101.

Hannigan, Laurie J., David M. Phillippo, Peter Hanlon, Laura Moss, Elaine W. Butterly, Neil Hawkins, Sofia Dias, Nicky J. Welton, and David A. McAllister. 2022. "Improving the Estimation of Subgroup Effects for Clinical Trial Participants with Multimorbidity by Incorporating Drug Class-Level Information in Bayesian Hierarchical Models: A Simulation Study." *Medical Decision Making* 42 (2): 228–40. https://doi.org/10.1177/0272989X211029556.

Khan, Muhammad Shahzeb, Izza Shahid, Tariq Jamal Siddiqi, Safi U. Khan, Haider J. Warraich, Stephen J. Greene, Javed Butler, and Erin D. Michos. 2020. "Ten-Year Trends in Enrollment of Women and Minorities in Pivotal Trials Supporting Recent US FDA Approval of Novel Cardiometabolic Drugs." *Journal of the American Heart Association* 9 (10): e015594. https://doi.org/10.1161/JAHA.119.015594.

Moore, Dirk F. 2016a. "Basic Concepts and Survival Distributions." In *Applied Survival Analysis Using r*. Springer. https://doi.org/10.1007/978-3-319-31245-3_2.

———. 2016b. "Prediction and Model Assessment." In *Applied Survival Analysis Using r*. Springer. https://doi.org/10.1007/978-3-319-31245-3_11.

Poon, Raymond, Khushbu Khanijow, Sabrina Umarjee, Ekaterina Fadiran, M. Yu, Lei Zhang, and Amish Parekh. 2013. "Participation of Women and Sex Analyses in Late-Phase Clinical Trials of New Molecular Entity Drugs and Biologics." *Journal of Women's Health* 22 (7): 604–16. https://doi.org/10.1089/jwh.2012.3753.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Schoenfeld, David. 1981. "The Asymptotic Properties of Nonparametric Tests for Comparing Survival Distributions." *Biometrika* 68 (1): 316–19. https://doi.org/10.1093/biomet/68.1.316.

Su, Shengli. 2023. *GLDreg: Fit GL Distribution to Data via Regression*. https://CRAN.R-project.org/package=GLDreg.

Therneau, Terry M. 2026. *A Package for Survival Analysis in r*. https://CRAN.R-project.org/package=survival.